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Optimizing cadmium (CD) alloy solar cells with sputtered copper-dopped zinc telluride (ZNTE:CU) back contacts in the presence of hydrogen

Abstract

A method of manufacturing a cadmium (Cd) alloy transmissive solar cell is provided. The method includes pumping a vacuum chamber to a base pressure and pumping the vacuum chamber to a sputtering pressure. The method includes providing into the vacuum chamber a first gas at a rate that balances a flow of the first gas in and out of the vacuum chamber with respect to the sputtering pressure and heating a surface of a partially manufactured cadmium (Cd) alloy transmissive solar cell within the vacuum chamber to a calibrated deposition temperature. The method includes providing into the vacuum chamber a second gas including at least a hydrogen gas (H.sub.2) at a proportional rate to achieve a target gas mix while maintaining the sputtering pressure and depositing a target material onto the surface to form a back contact section of the cadmium (Cd) alloy transmissive solar cell.

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Background/Summary

FIELD OF INVENTION

(1) The disclosure herein is related to manufacturing transmissive solar cells.

BACKGROUND

(2) Hydrogen (H) is a chemical element that, at standard conditions, is a gas of diatomic molecules with the formula H.sub.2 and called hydrogen gas, molecular hydrogen, or dihydrogen. Argon (Ar) is a chemical element that, at standard conditions, is stable gas of one atom and called argon gas. Oxygen (O) is a chemical element that, at standard conditions, is a gas of diatomic molecules with the formula O.sub.2 and called oxygen gas or dioxygen. Sputtering, generally, is an atomic/microscopic deposition of particles of a target material onto a surface of a substrate within a vacuum chamber containing a gas, such as argon gas (Ar). More particularly, conventional sputtering techniques are used to make extremely fine uniform layers of the target material (i.e., thin films) on the surface.

(3) Research and scholarship regarding the effects of the presence of hydrogen gas (H.sub.2) during sputtering of thin films in a vacuum chamber that includes argon gas (Ar) are minimal.

(4) Particularly, the effects of hydrogen gas (H.sub.2) on sputter deposited copper-doped zinc telluride (ZnTe:Cu) thin films was first explored at the National Renewable Energy Laboratory (NREL) in 1996 (see Gessert, Timothy A., A Fundamental Study of the Electrical and Optical Properties of Thin-Film, r.f.-Sputter-Deposited, Cu-Doped ZnTe (1996), herein referred to as “Thesis 1996”). Thesis 1996 shows that a resistivity of a film increased to an extent that the resistivity was unmeasurable, that no measurable changes in tellurium (Te), zinc (Zn) or copper (Cu) composition were found, and that a film transmissivity increased due to an increase in a bandgap for three concentrations of H.sub.2 (i.e., 1.28% hydrogen gas (H.sub.2), 0.2% hydrogen gas (H.sub.2) and 0.1% hydrogen gas (H.sub.2) in an argon gas (Ar)). (Thesis 1996 at pp. 80, PDF 87). Thesis 1996 provided no definitive explanation as to the increase in the bandgap for the sputter deposited copper-doped zinc telluride (ZnTe:Cu) thin films. Thesis 1996 theorized that hydrogen gas (H.sub.2) may have ionized and incorporated into the zinc telluride (ZnTe) lattice as a hydrogen-telluride (H—Te) bond. (Thesis 1996 at pp. 81, PDF 89). Thesis 1996 also theorized that the hydrogen gas (H.sub.2) may have reacted with residual oxygen gas (O.sub.2) in a vacuum chamber (i.e., creating a reducing environment) thereby reducing residual oxygen in the copper-doped zinc telluride (ZnTe:Cu) thin film. (Thesis 1996 at pp. 83, PDF 90). While Thesis 1996 showed positive changes in optical properties (i.e., increased film transmissivity), Thesis 1996 also indicated unfavorable changes in the electrical properties (i.e., increased film resistivity). Accordingly, because Thesis 1996 is limited to the study of thin films and teaches away from any application of creating electrical devices due to the unfavorable changes in the electrical properties, Thesis 1996 is not applicable to the different and more complex field of manufacturing transmissive solar cells (due in part to unexpected problems arising in the manufacturing transmissive solar cells that are not contemplated by the simpler study of thin films).

(5) In 2016, the effects of hydrogen gas (H.sub.2) on sputter deposited copper-doped zinc telluride (ZnTe:Cu) thin films was again explored. (see Faulkner, Brooke R., Variation of ZnTe:Cu Sputtering Target Preparation And Impacts On Film Properties and CdS/CdTe Solar Cell Performance (2016), herein referred to as “Faulkner 2016”). Faulkner 2016 shows a series of experiments that are similar to Thesis 1996 for sputter deposited copper-doped zinc telluride (ZnTe:Cu) thin films in concentrations of hydrogen gas (H.sub.2) in an argon gas (Ar) (i.e., 6.07% hydrogen gas (H.sub.2), 1.28% hydrogen gas (H.sub.2) and 0.77% hydrogen gas (H.sub.2)). (Faulkner 2016 at pp. 53, PDF 65). Faulkner 2016 shows that the bandgap increases as the hydrogen gas (H.sub.2) is added to the argon gas (Ar), which confirmed the results of Thesis 1996. (Faulkner 2016 at pp. 57. PDF 69). Faulkner 2016 theorizes that the hydrogen gas (H.sub.2) reduces the residual oxygen gas (O.sub.2) content in the copper-doped zinc telluride (ZnTe:Cu) thin film. Faulkner 2016 further presented results with no measurable oxygen gas (O.sub.2) in the argon gas (Ar) with the hydrogen gas (H.sub.2), which indicates that “other factors” (e.g., temperature of a substrate) may be contributing to the residual oxygen gas (O.sub.2) concentration. (Faulkner 2016 at pp. 55, PDF 67). Additionally, while Faulkner 2016 goes on to analyze the impact of the other factors on copper-doped zinc telluride (ZnTe:Cu), the analysis of Faulkner 2016 is limited to the research of the other factors as isolated parameters (e.g. temperature of a substrate without hydrogen gas (H.sub.2) variances in the argon gas (Ar)). Accordingly, because Faulkner 2016 is limited to the study of optical and structural properties of copper-doped zinc telluride (ZnTe:Cu) thin films and because Faulkner 2016 also indicates that hydrogen gas (H.sub.2) may have a negative impact on electrical properties of copper-doped zinc telluride (ZnTe:Cu) thin films (and thus would not be a good optimization for a CdTe solar cell back contact), Faulkner 2016 is also not applicable to the different and more complex field of manufacturing transmissive solar cells.

(6) Research and scholarship regarding the use of copper-doped zinc telluride (ZnTe:Cu) as a back contact for traditional cadmium telluride (CdTe) solar cell exists. However, because traditional cadmium telluride (CdTe) solar cells are three (3) microns (μm) to six (6) microns (μm) thick and inherently opaque, the research and scholarship regarding the use of

copper-doped zinc telluride (ZnTe:Cu) as the back contact for traditional cadmium telluride (CdTe) solar cells has focused on electrical and mechanical properties of cadmium telluride (CdTe), copper-doped zinc telluride (ZnTe:Cu), and other sections of the traditional cadmium telluride (CdTe) solar cells.
(7) Research and scholarship regarding the alternative fields of bifacial, tandems and transparent solar cells and optical properties thereof also exists. However, because this research and scholarship is centered on optical analysis and optimization of individual layers and materials of these solar cells without contemplation of electrical and mechanical properties, these alternative fields when applied to transmissive solar cells provide inconclusive data, unrelated teachings, and/or features that would destroy a function thereof. For example, optimization of the optical properties (which typically starts by reducing thickness, but is also greatly influenced by dopants, temperature, and crystal structure) comes at reduction of the electrical properties, leading to the incorrect conclusion that the optimization of one must come at the sacrifice of the other.
(8) Accordingly, because research and scholarship regarding traditional cadmium telluride (CdTe) solar cells and the alternative fields of bifacial, tandems and transparent solar cells is incongruent with transmissive solar cells, research and scholarship regarding traditional cadmium telluride (CdTe) solar cells and the alternative fields of bifacial, tandems and transparent solar cells are also not applicable to the different and complex field of manufacturing transmissive solar cells.
(9) There is a need for transmissive solar cells that provide a commercially viable photo electric performance. In turn, there is a need for optimization methods in manufacturing transmissive solar cells that contemplate bandgap, layer thickness, transmissivity, composition, and other properties, or any combination thereof, to arrive at commercially viable photo electric performance.
SUMMARY
(10) According to one or more embodiments, a method for manufacturing transmissive solar cells is provided.
(11) According to one or more embodiments, a method of manufacturing a cadmium (Cd) alloy transmissive solar cell is provided. The method includes pumping a vacuum chamber to a base pressure and admitting gas into the vacuum chamber to raise the base pressure to a sputtering pressure. The method includes providing into the vacuum chamber a first gas at a rate that balances a flow of the first gas in and out of the vacuum chamber with respect to the sputtering pressure. The method includes heating a surface of a partially manufactured cadmium (Cd) alloy transmissive solar cell within the vacuum chamber to a calibrated deposition temperature. The method includes providing into the vacuum chamber a second gas including at least a hydrogen gas (H.sub.2) at a proportional rate to achieve a target gas mix while maintaining the sputtering pressure. The method includes depositing a target material onto the surface to form a back contact section of the cadmium (Cd) alloy transmissive solar cell.
(12) According to one or more embodiments, a method of calibrating a deposition device to generate a plurality of time and temperature offsets is provided. The method includes setting a substrate holder system of the deposition device to a first set point temperature and continuously detecting a surface temperature on the glass surface until a normalization. The method includes recording a time and a surface temperature offset at the normalization. The method includes repeatedly incrementing the first set point temperature to a next set point temperature, detecting a next surface temperature at a next normalization respective to the next set point temperature, and recording a next time and a next surface temperature offset respective to the next set point temperature until the substrate holder system is set to a last set point temperature. The method includes generate the plurality of time and temperature offsets for the deposition device from each recording at each increment from the first set point temperature to the last set point temperature.
(13) According to one or more embodiments, any of the methods described herein can be implemented in devices, apparatuses, and systems, and with respect to one or more related methods.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) A more detailed understanding may be had from the following description, given by way of example in conjunction with the accompanying drawings, wherein like reference numerals in the figures indicate like elements, and wherein:
(2) FIG. **1** depicts an environment according to one or more embodiments;
(3) FIG. **2** depicts a transmissive cadmium (Cd) alloy solar cell according to one or more embodiments;
(4) FIG. **3** depicts an example deposition system and method according to one or more embodiments;
(5) FIG. **4** depicts a table and graphs according to one or more embodiments;
(6) FIG. **5** depicts a table and graphs according to one or more embodiments; and
(7) FIG. **6** depicts a graph according to one or more embodiments;
(8) FIG. **7** depicts a graph according to one or more embodiments;
(9) FIG. **8** depicts an example deposition system and method according to one or more embodiments; and
(10) FIG. **9** depicts a graph according to one or more embodiments.
DETAILED DESCRIPTION
(11) The disclosure herein relates to transmissive solar cells implemented in modules, devices, systems, windows, skylights, and buildings that provide commercially viable photo electric performances while having optimized properties. Thus, the disclosure herein relates to optimization methods in manufacturing these transmissive solar cells that contemplate bandgap, layer thickness, transmissivity, composition, and other properties, or any combination thereof, to

arrive at the commercially viable photo electric performances.

(12) According to one or more embodiments, an optimization method for depositing copper-doped zinc telluride (ZnTe:Cu) as a back contact for a cadmium (Cd) alloy transmissive alloy solar cell with a hydrogen concentration present in a deposition environment is provided. The deposition environment, generally, is a vacuum including argon gas (Ar). The hydrogen concentration include hydrogen gas (H.sub.2) at or between 0.01% to 0.10%, which is added to the deposition environment (i.e., the argon gas (Ar)) in contemplation of manufacturing the cadmium (Cd) alloy transmissive alloy solar cell. Thus, the optimization method includes the use of hydrogen gas (H.sub.2) during the deposition of the copper-doped zinc telluride (ZnTe:Cu) back contact on the cadmium (Cd) alloy transmissive alloy solar cell, in combination with changes with other deposition parameters to achieve both an increase in electrical properties and optical properties of the cadmium (Cd) alloy transmissive alloy solar cell.

(13) According to one or more embodiments, the optimization method creates a balanced optical and electrical cadmium (Cd) alloy transmissive alloy solar cell. That is, because the work of Thesis 1996 and Faulkner 2016 are limited in field and disclosure as discussed herein, the optimization method provides a mechanism to enhance both the optical and electrical properties of a copper-doped zinc telluride (ZnTe:Cu) back contact through incorporating hydrogen gas (H.sub.2) in argon gas (Ar). As discussed herein, different thicknesses of materials can be deposited for electrical and optical measurements. Thicker materials render better electrical measurements, however, due to the thickness, the thicker materials becomes inherently opaque and yields no optical measurements. Conversely, thinner material can provide optical measurements, yet yield no electrical measurements (i.e., too thin). Thus, because properly characterizing optical and electrical properties is extremely difficult, the optimization method focuses on an entirety of the cadmium (Cd) alloy transmissive alloy solar cell (not just one section) as well as electrical performance of the entire cadmium (Cd) alloy transmissive alloy solar cell (not just one section). Accordingly, the cadmium (Cd) alloy transmissive alloy solar cell and the optimization method herein provide a designed, deposited, and tested solar cell that is unrelated to simple single thin film scholarship and providing experimental results going well beyond any traditional/conventional technologies.

(14) Turning to FIG. **1**, an environment **100** is provided according to one or more embodiments. The environment **100** of FIG. **1** is shown oriented according to an X1-X2 axis, a Z1-Z2 axis, and a Y1-Y2 axis. The X1-X2 axis is generally horizontal as oriented in the Figures, with the X1-X2 axis having a direction between left (X1) and right (X2). Accordingly, the X1 direction is opposite the X2 direction, reference to a left side or left facing surface of a component may be referred to as an X1 side or an X1 surface of the component, and reference to a right side or right facing surface of a component may be referred to as an X2 side or an X2 surface of the component. The Z1-Z2 axis is generally vertical as oriented in the Figures, with the axis having a direction between down (Z1) and up (Z2). Accordingly, the Z1 direction is opposite the Z2 direction, reference to a lower or bottom side or a downwardly facing surface of a component may be referred to as a Z1 side or a Z1 surface, and reference to a top or upper side or upwardly facing surface of a component may be referred to as a Z2 side or a Z2 surface. Other orientations (e.g., tilted or angled orientations) can be made in accordance with the X1-X2 axis and Z1-Z2 axis. The Y1-Y2 axis is generally orthogonal to the X1-X2 axis and the Z1-Z2 axis and having a direction between forward (Y1) and back (Y2), though the Y1-Y2 is not shown orthogonal to the page of FIG. **1**. Accordingly, the Y1 direction is opposite the Y2 direction, reference to a forward side or forward facing surface of a component may be referred to as an Y1 side or an Y1 surface of the component, and reference to a back side or back facing surface of a component may be referred to as an Y2 side or an Y2 surface of the component.

(15) The environment **100** includes the sun **101** providing light energy or light **102** (e.g., from the Z2 direction). The sun **101** is representative of all light sources. The light **102** can be considered incident light or natural light, and the light **102** can represent light from sources other than the sun **102**. The light **102** can range across alight spectrum including, but not limited to, ultraviolet (UV) light, visible light, and infrared light.

(16) The environment **100** includes one or more sites, as represented by a building **105** (e.g., a skyscraper), a residence **107** (e.g., a house), and a surface area **109** (e.g., a field, a body of water, etc.). By way of example, the environment **100** and the one or more sites can include aquaculture areas (e.g., lake or pond where light transmission is beneficial to the water), agriculture areas (e.g., field or farm where light transmission is beneficial to the land), temporary structures (e.g., military forward medical units), structures needing power and daylight (e.g., remote machine sheds, electrical relay stations, pump stations, water monitoring, etc.), etc. Configured at the one or more sites are one or more solar devices **110** (e.g., high efficiency, bandgap, ultra-thin photovoltaic devices) that include one or more bandgap cadmium (Cd) alloy solar cells particularly designed to optimize a cadmium (Cd) bandgap and configure a transmissivity through composition, layer thickness, or both while maintaining commercially viable electrical performance, as described herein. That is, the one or more solar devices **110** have a power conversion efficiency (PCE) that provides commercially viable electrical performance. PCE, generally, is a measurement in percentage of an energy converted from the light **102** to electricity (e.g., a ratio of electrical power produced by a cell of the device **110** to the light **102** received).

(17) As shown in FIG. **1**, the one or more solar devices **110** can be implemented as building panels **110***a*, roof panels **110***b*, and/or racked panels **110***c*. According to one or more embodiments, the building panels **110***a* or the roof panels **110***b* can include solar devices **110** for use in independent or in mechanical stacking configurations for use in commercial and residential applications. According to one or more embodiments, the racked panels **110***c* can include solar devices **110** for use in independent or in mechanical stacking configurations for use in power plant applications. One or more technical effects, benefits, and advantages of the one or more solar devices **110** include balancing composition, thickness, and transmissivity, as well as maintaining/improving the total PCE and adding/improving anti-reflective coatings, in

contemplation of the building panels **110***a*, the roof panels **110***b*, and the racked panels **110***c* of the commercial, residential, and applications of FIG. **1**.

(18) Turning to FIG. **2**, a transmissive cadmium (Cd) alloy solar cell **200** is provided according to one or more embodiments. For ease of explanation, items and elements of the that are similar to previous Figures are reused and not reintroduced. The transmissive cadmium (Cd) alloy solar cell **200** can be representative of an example of single cell of a module including a plurality of transmissive cadmium (Cd) alloy solar cells **200**. Further, the transmissive cadmium (Cd) alloy solar cell **200** can be representative a cell of mechanically stacked solar modules within one or more apparatuses, devices, or systems. The transmissive cadmium (Cd) alloy solar cell **200** includes a ten percent (10%) or greater PCE, which can add to a total PCE of a mechanically stacked solar apparatus, device, or system. According to one or more embodiments, the transmissive cadmium (Cd) alloy solar cell **200** includes a fifteen percent (15%) or greater PCE, which can add to a total PCE of a mechanically stacked solar apparatus, device, or system. Accordingly, embodiments of the transmissive cadmium (Cd) alloy solar cell **200** include apparatuses, systems, and/or methods at any possible technical detail level of integration.

(19) The transmissive cadmium (Cd) alloy solar cell **200** of FIG. **2** is shown as a cross section oriented according to an X1-X2 axis and a Z1-Z2 axis. The X1-X2 axis is generally horizontal as oriented in the Figures, with the X1-X2 axis having a direction between left (X1) and right (X2). Accordingly, the X1 direction is opposite the X2 direction, reference to a left side or left facing surface of a component may be referred to as an X1 side or an X1 surface of the component, and reference to a right side or right facing surface of a component may be referred to as an X2 side or an X2 surface of the component. The Z1-Z2 axis is generally vertical as oriented in the Figures, with the axis having a direction between down (Z1) and up (Z2). Accordingly, the Z1 direction is opposite the Z2 direction, reference to a lower or bottom side or a downwardly facing surface of a component may be referred to as a Z1 side or a Z1 surface, and reference to a top or upper side or upwardly facing surface of a component may be referred to as a Z2 side or a Z2 surface. Other orientations (e.g., tilted or angled orientations) can be made in accordance with the X1-X2 axis and Z1-Z2 axis.

(20) The transmissive cadmium (Cd) alloy solar cell **200** receives, from at least the sun **101** (e.g., from the Z2 direction and at a Z2 side of the transmissive cadmium (Cd) alloy solar cell **200**), the light **102**. The transmissive cadmium (Cd) alloy solar cell **200** absorbs and converts one or more portions of the light **102** into electricity and transmits one or more remaining portions out of a Z1 side (e.g., a side opposite the sun **101**) of transmissive cadmium (Cd) alloy solar cell **200** and in the Z1 direction, as transmitted light **204**. The transmitted light **204** ranges across the light spectrum. The transmitted light **204** is at a lower intensity than the light **102** due to the absorption and conversion of the one or more portions of the light **102** by the transmissive cadmium (Cd) alloy solar cell **200**.

(21) The transmissive cadmium (Cd) alloy solar cell **200** includes a first substrate section **205**, a first conductive section **210**, a first transmissive absorber section **215**, and a second transmissive absorber section **220**. The transmissive cadmium (Cd) alloy solar cell **200** can further include a back contact section **225**, a second conductive section **230**, and/or a second substrate section **235**.

(22) According to one or more embodiments, a Z2 side (e.g., the sun side) of the first substrate section **205** faces outward from the transmissive cadmium (Cd) alloy solar cell **200** and is exposed to the light **102**, and a Z1 side (e.g., a side opposite the sun side) the first substrate section **205** faces inward (e.g., into to the transmissive cadmium (Cd) alloy solar cell **200**). The Z1 side of the first substrate section **205** is adjacent to a Z2 side of the first conductive section **210**. Adjacent in this context includes two sections being in contact and intimately bound. Adjacent can further include a blending of sections across the boundaries (such as a transfers of materials at an atomic level). A Z1 side of the first conductive section **210** is adjacent to a Z2 side of the first transmissive absorber section **215**. A Z1 side of the first transmissive absorber section **215** is adjacent to a Z2 side of the second transmissive absorber section **220**. A Z1 side of the second transmissive absorber section **220** is adjacent to a Z2 side of the back contact section **225**. A Z1 side of the back contact section **225** is adjacent to a Z2 side of the second conductive section **230**. A Z1 side of the second conductive section **230** is adjacent to a Z2 side of the second substrate section **235**. A Z2 side (e.g., a sun side) the second substrate section **235** faces inward (e.g., into to the transmissive cadmium (Cd) alloy solar cell **200**), and a Z1 side (e.g., a side opposite the sun side) of the second substrate section **235** faces outward from the transmissive cadmium (Cd) alloy solar cell **200**.

(23) Generally, the Z2 side of the first substrate section **205** receives the light **102**. The light **102** passes through, in the Z1 direction, the first substrate section **205**, the first conductive section **210**, and into the first transmissive absorber section **215** and the second transmissive absorber section **220**. The first transmissive absorber section **215** and the second transmissive absorber section **220** absorb (e.g., whether in combination or individually) one or more portions of the light **102** received from the first conductive section **210** and can transmit the one or more remaining portions, as the transmitted light **204**, towards the back contact section **225** (e.g., out of the Z1 side of the second transmissive absorber section **220** in the Z1 direction). The transmitted light **204** passes through, in the Z1 direction, the back contact section **225**, the second conductive section **230**, and the second substrate section **235** and exits a Z1 side of the second substrate section **235**.

(24) The transmissive cadmium (Cd) alloy solar cell **200** as a whole, along with each individual section or groups of sections, can be described by one or more properties. The one or more properties include, but are not limited to, a composition, a thickness, a transmissivity, and/or a bandgap. For ease of explanation and brevity, FIG. **2** shows example representations of these one or more properties in a particular section or across a particular group of section (though properties for all sections and combination of sections are contemplated herein).

(25) As shown in FIG. **2**, the transmissive cadmium (Cd) alloy solar cell **200** can include a total thickness **240**. The total thickness **240** can include at least a thickness **241** of the second transmissive absorber section **220**, a thickness **242** of the back contact section **225**, and a thickness **243** of the first transmissive absorber section **215**. Further, the thickness **241** and thickness **243** can be combined to provide a thickness **244** of the second transmissive absorber section and the first transmissive absorber section **215**. Furthermore, an internal thickness **245** can be a distance between the Z2 side of the first conductive section **210** and a Z1 side of the second conductive section **230**.
(26) As shown in FIG. **2**, the second transmissive absorber section **220**, the back contact section **225**, and the first transmissive absorber section **215** of the transmissive cadmium (Cd) alloy solar cell **200** can include compositions **251**, **252**, and **253**. Further, the second transmissive absorber section **220**, the back contact section **225**, and the first transmissive absorber section **215** of the transmissive cadmium (Cd) alloy solar cell **200** can include optimized bandgaps **261** (show as optimized bandgap **261***a* and optimized bandgap **261***b*), **262**, and **263**.
(27) The first substrate section **205** can be any glass or plastic that is transmissive. According to one or more embodiments, the transmissive quality of the first substrate section **205** includes a high transmissivity that transmits the light spectrum. According to one or more embodiments, the first substrate section **205** can include a section thickness along a range from two (2) millimeters (mm) to four (4) millimeters (mm) (e.g., at approximately 3.2 mm). The first substrate section **205** can be also be conductive.
(28) An example of the first substrate section **205** can include, but is not limited to, a fluorine-doped tin oxide (FTO) glass substrate. The FTO glass substrate is a transmissive (e.g., which includes being transparent) conductive substrate, distinctively coated with FTO to provide electrical conductivity. The FTO coating can be on the Z1 surface, the Z2 surface, or both of the first substrate section **205**.
(29) The first conductive section **210** can include a conductive glass, a conductive plastic, or other conductive material with a transmissive quality. The first conductive section **210** can include fluorine-doped tin oxide (FTO), indium-doped tin oxide (ITO), or sufficiently transparent and conductive layers oxide (e.g., a nanoparticle layer). According to one or more embodiments, the transmissive quality of the first conductive section **210** includes a transmissivity that transmits the light spectrum. The first conductive section **210** can be referred to as a transmissive conductive front contact or a front electrode. The substrate section **210** can be an n-type transparent material, for example, an ITO-coated glass or FTO-coated glass. The first conductive section **210** can include a section thickness along a range from two hundred (200) nanometers (nm) to six hundred (600) nanometers (nm) (e.g., at approximately 400 nm).
(30) The first transmissive absorber section **215** includes one or more properties, such as the composition **253**, the thickness **243**, a transmissivity, and/or the optimized bandgap **263**. The composition **253** of the first transmissive absorber section **215** can include a cadmium alloy and/or other transmissive absorber material. Examples of cadmium alloy include, but are not limited to, cadmium sulfide (CdS), cadmium selenide (CdSe), and cadmium telluride (CdTe), zinc telluride (ZnTe), and magnesium-doped zinc oxide (MZO). The first transmissive absorber section **215** can be an n-type semiconductor or a p-type semiconductor, or combination thereof. The composition **253** of the first transmissive absorber section **215** can be selected and configured in view of the description herein.
(31) The thickness **243** of the first transmissive absorber section **215** can include a one hundred (100) or less nanometer (nm) section thickness. The thickness **243** of the first transmissive absorber section **215** can include a section thickness along a range from fifty (50) nanometers (nm) to one hundred fifty (150) nanometers (nm) (e.g., at approximately 200 nanometers (nm)). As shown in FIG. **2** optimized bandgap **263** of the first transmissive absorber section **215** can include be greater than 2.0 eV.
(32) The second transmissive absorber section **220** is transmissive. The second transmissive absorber section **220** includes one or more properties, such as the composition **251**, the thickness **241**, a transmissivity, and/or the optimized bandgap **261**. The composition **251** of the second transmissive absorber section **220** includes a cadmium alloy and/or other transmissive absorber material with an optimized bandgap (e.g., the optimized bandgap **261**).
(33) More particularly, the second transmissive absorber section **220** balances the composition **251**, the thickness **241**, and/or a transmissivity to provide the optimized bandgap **261** (i.e., a high bandgap). The composition **251** can correspond to the materials, material concentrations, bandgap, and other physical, electrical, and crystalline properties of the one or more sections of the one or more transmissive cadmium (Cd) alloy solar cells **200**. Further, to design the optimized bandgap **261** for any irradiance wavelength ranges, the individual sections of the transmissive cadmium (Cd) alloy solar cell **200** can be configured with compositions that assist with, enhance, or do not interfere with achieving the optimized bandgap **261**.
(34) Examples of the composition **251** (e.g., cadmium alloys) include, but are not limited to, cadmium telluride (CdTe), cadmium selenide telluride (CdSeTe), and cadmium selenide (CdSe).
(35) The composition **251** (e.g., of the second transmissive absorber section **220**, as well as other sections of the transmissive cadmium (Cd) alloy solar cell **200**) can include, but is not limited to, cadmium, zinc, selenium, tin, oxygen, copper, nitrogen, aluminum, carbon, gold, platinum, palladium, sulfur, and other additives. By further configuring the composition **251**, the second transmissive absorber section **220** can be an n-type semiconductor, an i-type semiconductor, a p-type semiconductor, or combination thereof. According to one or more embodiments, the second transmissive absorber section **220** can have changing ratios n-type semiconductor, i-type semiconductor, and p-type semiconductor.
(36) The second transmissive absorber section **220** includes the thickness **241** (e.g., a section thickness). The thickness **241** can be a width of the second transmissive absorber section **220** from the Z2 surface of the second transmissive

absorber section **220** to the Z1 surface of the second transmissive absorber section **220**. According to one or more embodiments, the thickness **241** can be at or less than seven hundred (700) nanometers (nm), for example, at or less than six hundred fifty (650) nanometer (nm) or at or less than five hundred (500) nanometers (nm). According to one or more embodiments, the thickness **241** can be along a range of one hundred (100) nanometers (nm) to seven hundred (700) nanometers (nm), for example, along a range from one hundred fifty (150) nanometers (nm) to six hundred fifty (650) nanometers (nm) (e.g., at approximately 500 nm, 300 nm, or 200 nm). The thickness **241** can be one hundred (100) nanometers (nm), one hundred fifty (150) nanometers (nm), two hundred (200) nanometers (nm), two hundred fifty (250) nanometers (nm), three hundred (300) nanometers (nm), three hundred fifty (350) nanometers (nm), four hundred (400) nanometers (nm), etc.

(37) The composition **251** and the thickness **241** can contribute to a transmissivity **260** of the transmissive cadmium (Cd) alloy solar cell **200**, but are not the only factors. The transmissivity **260** is an ability of the transmissive cadmium (Cd) alloy solar cell **200** to permit one or more portions of the light spectrum (e.g., irradiance of ultraviolet (UV) light, visible light, and infrared light) that is not absorbed to transmit therethrough. The transmissivity **260** of the transmissive cadmium (Cd) alloy solar cell **200** can be designed for portions of the light spectrum irrespective of the other portions.

(38) According to one or more embodiments, the transmissivity **260** of the transmissive cadmium (Cd) alloy solar cell **200** includes at least ten percent (10%) transmissivity for one or more portions of a first irradiance wavelength range. The first irradiance wavelength range can include a wavelength range less than and equal or approximate to the wavelength of the optimized bandgap **261**. The optimized bandgap **261** can be a slope, a curve, or other shape along a second irradiance wavelength. The second irradiance wavelength range can include a wavelength range equal or approximate to a beginning and end of an interior slope of the wavelength of the optimized bandgap **261**. The transmissivity **260** of the transmissive cadmium (Cd) alloy solar cell **200** includes at least sixty-five percent (65%) transmissivity for one or more portions of a third irradiance wavelength range. The third irradiance wavelength range can include a wavelength range greater than and equal or approximate to the wavelength of the optimized bandgap **261**. The first, second, and third irradiance wavelength ranges can overlap. According to one or more embodiments, the transmissivity **260** of the transmissive cadmium (Cd) alloy solar cell **200** includes at least fifteen percent (15%) or greater for the one or more portions of the first irradiance wavelength range and includes at least seventy percent (70%) or greater for the one or more portions of the third irradiance wavelength range.

(39) Thus, a plurality of compositions of the transmissive cadmium (Cd) alloy solar cell **200** (e.g., the composition **251**), a plurality of thicknesses of the sections (e.g., the thickness **241**), and a plurality of thicknesses of groups of sections (e.g., a combined thickness **244** or an internal thickness **245**) are contemplated by the transmissive cadmium (Cd) alloy solar cell **200** to achieve a high bandgap (e.g., the optimized bandgap **261**) and relative transmissive qualities (e.g., the transmissivity **260**) for the transmissive cadmium (Cd) alloy solar cell **200**.

(40) The combined thickness **244** can be an addition of section thicknesses for the first and second transmissive absorber sections **215** and **220**. The combined thickness **244** can be a width from the Z2 surface of the first transmissive absorber section **215** to the Z1 surface of the second transmissive absorber section **220**. According to one or more embodiments, the combined thickness **244** can be at or less than one thousand (1000) nanometers (nm), for example, at or less than seven hundred fifty (750) nanometer (nm).

(41) The internal thickness **245** can be a width from the Z2 surface of the first conductive section **210** to the Z1 surface of the second conductive section **230**. According to one or more embodiments, the total thickness can be less than two thousand (2,000) nanometers (nm), for example, along a range of five hundred (500) nanometers (nm) to one thousand five hundred (1,500) nanometers (nm).

(42) The back contact section **225** is transmissive. The back contact section **225** includes one or more properties, such as the composition **252**, the thickness **242**, a transmissivity, and/or the optimized bandgap **262**. The composition **252** of the back contact section **225** can include can include one or more conductors. The composition **252** of the back contact section **225** can include one or more semiconductors, which are different than conductors and insulators. According to one or more embodiments, the back contact section **225** provides a tunnel junction or an ohmic contact on a bottom, back, rear or Z1 side of the second transmissive absorber section **220**. According to one or more embodiments, the back contact section **225** can have no bias on voltage or current changes and provide a stable and low resistance contact for cadmium (Cd) alloys.

(43) For instance, the composition **252** of the back contact section **225** can include one or more gold (Au), aluminum (Al), platinum (Pt), zinc telluride (ZnTe) (e.g., doped with Copper (Cu) or other material), zinc telluride (ZnTe) based alloy (e.g., doped with Copper (Cu) or other material), or other material/alloy at a thickness that maintains transmissivity. The composition **252** of the back contact section **225** can be selected and configured in view of the description herein.

(44) The thickness **242** of the back contact section **225** can include a section thickness along a range from fifty (50) nanometers (nm) to one hundred fifty (150) nanometers (nm) (e.g., at approximately 200 nm of back contact. Note that conductive sections of conventional and traditional solar cells are generally opaque; thus, the transmissive cadmium (Cd) alloy solar cell **200** either replaces these conductive sections with transmissive conductors and/or semiconductors or reduces a thickness thereof to a transmissive dimension (i.e., five (5) to two hundred (200) nanometers or nm for gold (Au)).

(45) As shown in FIG. **2** optimized bandgap **262** of the back contact section **225** can include be greater than 2.0 eV.

(46) The second conductive section **230** is transmissive. The second conductive section **230** (e.g., a back electrode) can

include material that provides an electrical path for electricity to leave the transmissive cadmium (Cd) alloy solar cell **200**. The material of the second conductive section **230** can include a conductive glass, a conductive plastic, or other conductive material with a transmissive quality (e.g., including FTO, ITO, or other transparent conductive oxide, as well as or in the alternative one or more of ultra-thin carbon (C), ultra-thin aluminum (Al), ultra-thin gold (Au), and graphene). The second conductive section **225** can include a section thickness along a range from two hundred (200) nanometers (nm) to six hundred (600) nanometers (nm) (e.g., at approximately 400 nm).

(47) The second substrate section **235** can be any glass or plastic that is transmissive. According to one or more embodiments, the transmissive quality of the second substrate section **235** includes a high transmissivity that transmits the light spectrum. The second substrate section **235** can be also be conductive. According to one or more embodiments, the second substrate section **235** can include a section thickness along a range from one (1) millimeters (mm) to four (4) millimeters (mm) (e.g., at approximately 3.2 mm). An example of the second substrate section **235** can include, but is not limited to, a FTO-coated glass of ITO-coated glass.

(48) As described herein, the transmissive cadmium (Cd) alloy solar cell **200** receives the light **102** from at least the sun **101**. As shown in FIG. **2**, the light **102** can include at least one or more portions **291.03**, **291.04**, **291.05**, **291.06**, **291.07**, **291.08**, **291.09**, **291.10**, **291.11**, and **291.12**. The one or more portions **291.03**, **291.04**, **291.05**, **291.06**, **291.07**, **291.08**, **291.09**, **291.10**, **291.11**, and **291.12** can respectively represent the light **102** at approximately three hundred (300), four hundred (500), five hundred (500), six hundred (600), seven hundred (700), eight hundred (800), nine hundred (900), one thousand (1,000), one thousand one hundred (1,100), and one thousand two hundred (1,200) nanometers (nm) wavelengths. The one or more portions **291.03**, **291.04**, **291.05**, **291.06**, **291.07**, **291.08**, **291.09**, **291.10**, **291.11**, and **291.12** are depicted as solid arrows to represent that the light **102** at that wavelength has a full amount of energy. Note that these wavelength values illustrate the light **102** as presented by the sun **101** across the full light spectrum including between these values and not at just those exact values.

(49) The transmissive cadmium (Cd) alloy solar cell **200** absorbs and converts one or more portions **291.03**, **291.04**, **291.05**, **291.06**, **291.07**, **291.08**, **291.09**, **291.10**, **291.11**, and **291.12** into electrical power or electricity and transmits one or more remaining portions **296.03**, **296.04**, **296.05**, **296.06**, **296.07**, **296.08**, **296.09**, **296.10**, **296.11**, and **296.12** out of the Z1 side (e.g., a side opposite the sun **101**) of transmissive cadmium (Cd) alloy solar cell **200** and in the Z1 direction, as the transmitted light **204**. Similarly, the one or more remaining portions **296.03**, **296.04**, **296.05**, **296.06**, **296.07**, **296.08**, **296.09**, **296.10**, **296.11**, and **296.12** can respectively represent the light **204** at approximately three hundred (300), four hundred (500), five hundred (500), six hundred (600), seven hundred (700), eight hundred (800), nine hundred (900), one thousand (1,000), one thousand one hundred (1,100), and one thousand two hundred (1,200) nanometers (nm) wavelengths. Note that these wavelength values illustrate the light **204** as affected by the transmissive cadmium (Cd) alloy solar cell **200** across the full light spectrum including between these values and not at just those exact values. In this regard, the one or more remaining portions **296.03**, **296.04**, **296.05**, **296.06**, **296.07**, **296.08**, **296.09**, **296.10**, **296.11**, and **296.12** are depicted with arrows that include degrees of shading to illustrate a spectral response of the second transmissive absorber section **220** to the one or more portions **291.03**, **291.04**, **291.05**, **291.06**, **291.07**, **291.08**, **291.09**, **291.10**, **291.11**, and **291.12**. When increased or optimized, the optimized bandgap **261***a* is shown with respect to the light portions **291.08** and remaining light portion **296.08** to represent an increased bandgap value of 1.70 eV. When further increased or further optimized, the optimized bandgap **261***b* is shown with respect to the light portions **291.06** and remaining light portion **296.06** to represent an increased bandgap value of 2.00 eV.

(50) The sections of the transmissive cadmium (Cd) alloy solar cell **200** are manufactured in a vertical or Z direction, and each Z surface of the sections of the transmissive cadmium (Cd) alloy solar cell **200** can include an anti-reflective coating. The antireflective coatings can be manufactured at particular thicknesses, and tuning for every section of the transmissive cadmium (Cd) alloy solar cell **200**.

(51) According to one or more embodiments, the transmissive cadmium (Cd) alloy solar cell **200** can be manufactured in a superstrate configuration in an order where the layers are deposited in decreasing deposition temperatures. For instance, to achieve the optimized bandgap **261** as described herein, deposition of the sections can be from section **235** to section **205** with decreasing deposition temperatures, so the deposition of any subsequent sections does not detrimentally affect one or more proceeding sections. Further, in the superstrate configuration, the section are deposited on a “superstrate” (e.g., usually a transparent/transmissive material like glass) so that the light **102** enters through the superstrate, which is the first section deposited.

(52) According to one or more embodiments, the transmissive cadmium (Cd) alloy solar cell **200** can be manufactured in a substrate configuration in an order where the layers are deposited in decreasing deposition temperatures. For instance, to achieve the optimized bandgap **261** as described herein, deposition of the sections can be from section **205** to section **235** with decreasing deposition temperatures, so the deposition of any subsequent sections does not detrimentally affect one or more proceeding sections. Further, in the superstrate configuration, the section are deposited on a “substrate” (e.g., usually a rigid material like glass or metal) so that the light **102** enters through a top section, which is the last section deposited.

(53) FIG. **3** depicts an example deposition system **300** and method **301** according to one or more embodiments. The example deposition system **300** can be any deposition system, example of which include sputtering systems, evaporation systems, sublimation systems, chemical vapor deposition systems, magnetron sputtering systems, atomic layer depositions systems, pulsed laser deposition systems, etc. The method **301** is an example operational process of the

example deposition system **300**. For ease of explanation, the example deposition system **300** and the method **301** are described with respect to a sputtering operation, though not limited thereto.
(54) The example deposition system **300** includes a vacuum chamber **303**. The vacuum chamber **303** is a sealed chamber where the sputtering operation occurs and a low pressure is maintained to facilitate a movement of ions. A target material **304** (e.g., a cathode) is mounted within the vacuum chamber **303** and is a material to be deposited. The substrate **307** (e.g., an anode) includes a surface **306** where the target material **304** will be deposited, and grounded or maintained at a potential to provide material benefits to the deposition. For instance, with respect to sputtering, after the target material **304** and the substrate **307** are configured in the vacuum chamber **303**, the vacuum chamber **303** is pressurized and the first gas **317** is introduced therein. The target material **304** within the vacuum chamber **303** is provided from a sputter gun **305** to coat (e.g., to be deposited on) the surface **306** of the substrate **307**. More particularly, when a voltage is applied to the target material **304** via the sputter gun **305**, ions of the first gas **317** are accelerated towards the target material **304** and knock off atoms therefrom. These atoms are then deposited on the surface **306** of the substrate **307**, forming a section thereon. Thus, any section of the substrate **307** can be formed by sputtering (e.g., ionization, ion bombardment, and deposition).
(55) The method **301** refers to sputtering sections of the transmissive cadmium (Cd) alloy solar cell **200** of FIG. **2** according to one or more embodiments. According to one or more embodiments, materials deposited in a superstrate configuration from a sun side to a side opposite the sun side to sequentially form sections of the transmissive cadmium (Cd) alloy solar cell **200** of FIG. **2**. According to one or more other embodiments, the materials can be sequentially deposit in a substrate configuration from the side opposite the sun side to the sun side.
(56) The example deposition system **300** also includes shutters **308** and **309**, a substrate holder system **310**, a pumping system **312**, and mass flow controllers **313** and **315** that respectively provide a first gas **317** and a second gas **319**. The shutters **308** and **309** can be panels that are movable in and out of position between the target material **304** and the substrate to respectively cover and expose both, as well as intermediate positions. The substrate holder system can include a heating device that based on a set point temperature heats the substrate in preparation for deposition. The pumping system **312** draws gas particles from the vacuum chamber **303** to leave behind a partial vacuum. The pumping system **312** is operated to provide at least a base pressure and a sputtering pressure within the vacuum chamber **303**. The mass flow controllers **313** and **315** are devices that accurately control a supply the first gas **317** and the second gas **319**, subsequently. The first gas **317** can be an inert gas (e.g., argon gas (Ar)) introduced into the vacuum chamber **303**, which is ionized to create a plasma. The second gas **319** can be hydrogen gas (H.sub.2) though other gases and gas mixtures are contemplated. Note that while one item is shown in the example deposition system **300**, any individual item can represent more than one of that item.
(57) The method **301** begins at block **330**, where one or more substrate section materials are deposited. By way of example, after the one or more substrate section materials are placed on the sputter gun **305** and the substrate **307** is mounted within the vacuum chamber **303**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controller **313** introduces the first gas **317**. The substrate holder system **310** heats the substrate **307**. Then, the one or more substrate section materials are sputtered onto the substrate **307** (e.g., to complete the substrate section **205**).
(58) At block **340**, one or more conductive section materials are deposited. Similarly, and by way of example, after the one or more conductive section materials are placed on the sputter gun **305**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controller **313** introduces the first gas **317**. The substrate holder system **310** heats the substrate **307**. Then, the one or more conductive section materials are sputtered onto the substrate **307** (e.g., onto the substrate section **205** to complete the conductive section **210**).
(59) At block **350**, one or more first transmissive absorber section materials are deposited. The one or more first transmissive absorber section materials can include MZO, cadmium (Cd), selenide (Se), telluride (Te), or any combination thereof. According to one or more embodiments, the one or more first transmissive absorber section materials include cadmium selenide telluride (CdSeTe) with a selenium concentration of fifty percent (50%) or more. The selenium concentration can be at or approximately ninety percent (90%). Accordingly, after the one or more first transmissive absorber section materials are placed on the sputter gun **305**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controller **313** introduces the first gas **317**.
(60) The substrate holder system **310** heats the substrate **307**. The substrate holder system **310** is set to a set point temperature. For example, the substrate holder system **310** is set to a set point temperature to heat a deposition surface of a partially manufactured cadmium (Cd) alloy transmissive to a calibrated first transmissive absorber deposition temperature (e.g., a surface of the conductive section **210**). The calibrated first transmissive absorber deposition temperature is a surface offset temperature of the substrate **307** that differs from the set point temperature. The calibrated first transmissive absorber deposition temperature is required to achieve a temperature on the surface (for receiving the one or more first transmissive absorber section materials) while guaranteeing integrity of the first transmissive absorber section during subsequent depositions of other sections. The calibrated first transmissive absorber deposition temperature can be a value along a range of two hundred (200) degrees Celsius (° C.) to six hundred (600) degrees Celsius (° C.). Two sputter guns **305** can be used to deposit a MZO followed by a cadmium selenide telluride (CdSeTe) to form the first transmissive absorber section. In this regard, a first sputter gun **305** can sputter the MZO (e.g., onto a surface of the conductive section **210**), which has a calibrated first transmissive absorber deposition temperature of three hundred (300) degrees Celsius (° C.). Then, a second sputter gun **305** can sputter the cadmium selenide telluride (CdSeTe) onto the

MZO, which can have a same calibrated first transmissive absorber deposition temperature of three hundred (300) degrees Celsius (° C.). Once sputtering of the cadmium selenide telluride (CdSeTe) is complete, the first transmissive absorber section is formed (e.g., the first transmissive absorber section **215**).

(61) At block **360**, one or more second transmissive absorber section materials are deposited. The one or more second transmissive absorber section materials can include cadmium (Cd), selenide (Se), telluride (Te), or any combination thereof. According to one or more embodiments, the one or more second transmissive absorber section materials include a cadmium (Cd) alloy, e.g., cadmium selenide telluride (CdSeTe), cadmium selenide (CdSe), cadmium telluride (CdTe), or a cadmium selenide telluride/cadmium telluride (CdSeTe/CdTe) combination. Accordingly, after the one or more second transmissive absorber section materials are placed on the sputter gun **305**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controller **313** introduces the first gas **317**.

(62) The substrate holder system **310** heats the substrate **307**. The substrate holder system **310** is set to a set point temperature. For example, the substrate holder system **310** is set to a set point temperature to heat a deposition surface of a partially manufactured cadmium (Cd) alloy transmissive to an calibrated absorber temperature (e.g., a surface of the first transmissive absorber section **215**). The calibrated second transmissive absorber temperature is a surface offset temperature of the substrate **307** that differs from the set point temperature. The calibrated second transmissive absorber temperature is required to achieve a temperature on the surface (for receiving the one or more second transmissive absorber section materials) while guaranteeing integrity of the second transmissive absorber section during subsequent depositions of other sections. The calibrated second transmissive absorber temperature can be along a range of one hundred (100) degrees Celsius (° C.) to three hundred fifty (350) degrees Celsius (° C.). The calibrated second transmissive absorber temperature can be at or approximately two hundred ten (210) degrees Celsius (° C.) or two hundred twenty (220) degrees Celsius (° C.). One or more technical effects, advantages, and benefits of the method **301** include, but are not limited to, the calibrated second transmissive absorber temperature being lower (a cooler temperature) than the calibrated first transmissive absorber deposition temperature so that the first transmissive absorber section or any preceding section is not damaged or destroyed during deposition of the one or more absorber second transmissive absorber materials. Further, the calibrated second transmissive absorber temperature being lower (a cooler temperature) than the calibrated first transmissive absorber deposition temperature solves the technical problem of the selenium (Se) reevaporates and leaves the substrate **307**, thereby preventing the manufacturing of a working transmissive cadmium (Cd) alloy solar cells.

(63) Additionally, at block **360**, a post deposition annealing can be performed. The post deposition annealing can be applied to the one or more second transmissive absorber section materials. The post deposition annealing can be applied for an annealing time period and at an annealing temperature. The annealing time period can be at or approximately one (1) hour. The annealing temperature can be a surface offset temperature along a range of one hundred (100) degrees Celsius (° C.) to three hundred fifty (350) degrees Celsius (° C.). The annealing temperature can be at or approximately two hundred ten (210) degrees Celsius (° C.) or two hundred twenty (220) degrees Celsius (° C.).

(64) Additionally, at block **360**, a cadmium chloride (CdCl2) post-deposition treatment can be performed. The cadmium chloride (CdCl2) post-deposition treatment can be applied to the one or more second transmissive absorber section materials at a treatment temperature. The treatment temperature is also a surface offset temperature of the substrate **307** that differs from the set point temperature. The treatment temperature can be a value along a range of three hundred eighty-five (385) degrees Celsius (° C.) to four hundred twenty-five (425) degrees Celsius (° C.). The treatment temperature can be at or approximately four hundred (400) degrees Celsius (° C.).

(65) At block **370**, a back contact section is manufactured. Accordingly, after the target material **304** is placed on the sputter gun **305**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controllers **313** and **315** introduce the first gas **317** and the second gas **319**, respectively.

(66) At sub-block **371**, the vacuum chamber **303** is pumped by the pumping system **312** to a base pressure. The base pressure can be at or lower than 1e-5 Torr. The base pressure can be at or approximately 9e-8 Torr. One or more technical effects, advantages, and benefits of the method **301** include, but are not limited to, the base pressure removing contaminates and residual oxygen from the vacuum chamber thereby effectively cleaning therein.

(67) At sub-block **373**, the vacuum chamber **303** is pumped by the pumping system **312** to a sputtering pressure. For example, the vacuum chamber **303** is raised from the base pressure to the sputtering pressure. The sputtering pressure can be at or lower than 10e-3 Torr. The sputtering pressure can be at or approximately 3e-3 Torr.

(68) At sub-block **375**, the mass flow controller **313** provides the first gas **317** into the vacuum chamber **303**. The first gas **317** can be provided at a rate that balances a flow of the first gas **317** in and out of the vacuum chamber **303** with respect to the sputtering pressure. For example, the mass flow controller **313** and pumping system **312** can work in concert to achieve the sputtering pressure.

(69) At sub-block **377**, the substrate holder system **310** heats the substrate **307**. The substrate holder system **310** can be set to a set point temperature to heat the substrate **307** until the surface **306** reaches a calibrated deposition temperature. The calibrated deposition temperature is a surface offset temperature of the substrate **307** that differs from the set point temperature. The calibrated deposition temperature is required to achieve a back contact section deposition temperature on the surface **306** (for receiving the one or more back contact section materials) while guaranteeing integrity of all preceding section during back contact deposition and guaranteeing integrity of the back contact section during subsequent depositions of other sections. The calibrated deposition temperature can be along a range of two hundred (200) degrees

Celsius (° C.) to six hundred (600) degrees Celsius (° C.). The calibrated deposition temperature can be at or approximately three hundred (300) degrees Celsius (° C.). One or more technical effects, advantages, and benefits of the method **301** include, but are not limited to, the calibrated absorber temperature being lower (a cooler temperature) than the calibrated first transmissive absorber deposition temperature so that the first transmissive absorber section or any preceding section is not damaged or destroyed during deposition of the one or more second transmissive absorber section materials.

(70) At sub-block **378**, the mass flow controller **315** provides the second gas **319** into the vacuum chamber **303**. The second gas **319** can be provided at a rate that balances a flow of the first gas **317** and the second gas **319** in and out of the vacuum chamber **303** with respect to the sputtering pressure. For example, the mass flow controllers **313** and **315** and pumping system **312** can work in concert to achieve/maintain the sputtering pressure. According to one or more embodiments, the mass flow controllers **313** and **315** and pumping system **312** respectively provide into the vacuum chamber **303** the first gas **317** and the second gas **319** at a proportional rate to achieve a target gas mix while maintaining the sputtering pressure (i.e., while balancing the argon gas (Ar) and the hydrogen gas (H.sub.2) and while running the pumping system **312**). The target gas mix can include the hydrogen gas (H.sub.2) at or below 1.00%. The target gas mix can include the hydrogen gas (H.sub.2) at or approximately at 0.1%. According to one or more embodiments, the first gas **317** and the second gas **319** are provided across a plurality of phases to achieve the target gas mix. According to one or more embodiments, the hydrogen gas (H.sub.2) includes a preblended hydrogen gas mix (e.g., for multistage gas mixing).

(71) At sub-block **379**, the target material **304** is deposited. The target material **304** can include zinc (Zn), telluride (Te), or any combination thereof. According to one or more embodiments, the target material **304** can include copper-doped zinc telluride (ZnTe:Cu). The target material **304** is deposited onto the surface **306** to form a back contact section of the cadmium (Cd) alloy transmissive solar cell. The deposition of the target material **304** is controlled by operating the shutters **308** and **309**. The deposition of the target material **304** is controlled by operating the shutters **308** and **309** while adjusting the target gas mix.

(72) Additionally, at sub-block **379**, the target material **304** is deposited onto the surface **306** at a deposition rate. The deposition rate of the target material **304** is controlled by setting a power of the sputter gun **305**. The power of the sputter gun **305** can be set to a calibrated power density. The deposition rate of the target material **304** is also controlled by performing the depositing for a specified time. The deposition rate of the target material **304** can be one (1) to five (5) angstroms per second. The calibrated power density and the specified time can together achieve a section thickness. The section thickness can be along a range of twenty (20) nanometers (nm) to one hundred and fifty (150) nanometers (nm). Thus, the back contact section is deposited.

(73) Additionally, at sub-block **379**, the target material **304** can be deposited in phases.

(74) According to one or more embodiments, a pre-sputter phase can be implemented. The pre-sputter phase can be for at or approximately five (5) minutes or until normal the vacuum chamber has normalized/stabilized. The hydrogen gas (H.sub.2) is turned off during the pre-putter phase, e.g., at the beginning of the pre-sputter phase. During the pre-sputter phase, the target material **304** is turned on (e.g., electricity is being applied to produce a plasma), though there in no depositing on the surface **306** because one or both of the shutters **308** and **309** are in a closed position. Once normalized/stabilizes, then the shutters **308** and **309** are opened. By way of example, open the shutter **308** over the target material **305** and turn on plasma and sputtering materials into the vacuum chamber **303** for a period of time until plasma and environment stabilizes, and then open the shutter **309** over the substrate **307**.

(75) According to one or more embodiments, a flow of the hydrogen gas (H.sub.2) is turn off partway through deposition (e.g., deposit one (1) nanometers (nm) to twenty (20) nanometers (nm) of the back contact layer section, then turn off the hydrogen gas (H.sub.2)).

(76) According to one or more embodiments, a flow of the hydrogen gas (H.sub.2) is turn on partway through deposition (e.g., deposit one (1) nanometers (nm) to twenty (20) nanometers (nm) of the back contact layer section, then turn on the hydrogen gas (H.sub.2)).

(77) Generally, the hydrogen gas (H.sub.2) and related plasma-generated species (e.g., H, H.sub.2, H.sub.3, etc.) can interact with residual oxygen and related plasma-generated species (e.g., O, O.sub.2, O.sub.3) and water and related plasma-generated species (e.g., H.sub.2O, H, H.sub.2, H.sub.3, O, O.sub.2, O.sub.3, etc.), producing species (e.g., OH, H.sub.2O, etc.) that are less likely than other species (e.g., O, O.sub.2) to incorporate into the zinc telluride-based (ZnTe) contact materials. For instance, the hydrogen gas (H.sub.2) may not go into the back contact section, may bond with residual oxygen (e.g., OH), may make water (H.sub.2O) and/or take other forms (e.g., H, H.sub.2, O.sub.3), and may get pumped out of the vacuum chamber **303** resulting in an environment that is less likely than other species (e.g., O, O.sub.2) to incorporate into the zinc telluride-based (ZnTe) contact materials.

(78) At block **380**, one or more back electrode section materials are deposited. Similarly, and by way of example, after the one or more back electrode section materials are placed on the sputter gun **305**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controller **313** introduces the first gas **317**. The substrate holder system **310** heats the substrate **307**. Then, the one or more back electrode section materials are sputtered onto the substrate **307** (e.g., onto the back contact section **225** to complete the back electrode section **230**).

(79) At block **390**, one or more back substrate section materials are deposited. Similarly, and by way of example, after the one or more back substrate section materials are placed on the sputter gun **305**, the pumping system **312** pressurizes the vacuum chamber **303** and the mass flow controller **313** introduces the first gas **317**. The substrate holder system **310** heats

the substrate **307**. Then, the one or more back substrate section materials are sputtered onto the substrate **307** (e.g., onto the back electrode section **230** to complete the back substrate section **235**). Thus, the transmissive cadmium (Cd) alloy solar cell **200** of FIG. **2** is complete.

(80) FIG. **4** depicts a table **400** and graphs **401** and **402**. The table **400** and graphs **401** and **402** provide hydrogen gas (H.sub.2) research with respect to a particularly set of lab experiments on a particular set of cell samples (e.g., Group 29 samples, as labeled in the real world lab). The table **400** provides a baseline (row **411**) with zero percent (0%) hydrogen gas (H.sub.2), first target gas mix (row **413**) with point one percent (0.1%) hydrogen gas (H.sub.2), and second target gas mix (row **415**) with one percent (1.0%) hydrogen gas (H.sub.2). The graph **401** depicts average PCE data (y-axis **420**) via diagrams **431**, **433**, **435** respective to the table **400** for the three rows **411**, **413**, **415** of table **400**. The graph **402** depicts average voltage data (y-axis **420**) via diagrams **451**, **453**, **455** respective to the table **400** for the three rows **411**, **412**, **413** of table **400**.

(81) The Group 29 samples used a traditional transmissive device architecture to create a baseline (see row **411** of table **400**) of an impact on the copper-doped zinc telluride (ZnTe:Cu) back contact section within an already understood and quantified transmissive device architecture. Further, the table **400** and the graphs **401** and **402** depicts the results of Group 29 samples, where the presence of hydrogen gas (H.sub.2) (see rows **413** and **415** of table **400**) significantly reduced a corresponding device performance (expected based on Thesis 1996 original resistivity findings) and also reduced a corresponding total device transmissivity (unexpected based on Thesis 1996 original optical findings).

(82) FIG. **5** depicts a table **500** and graphs **501** and **502** according to one or more embodiments. The table **500** and graphs **501** and **502** repeat the Group 29 experiment series with Group 39 samples. As compared to the Group 29 samples, the copper-doped zinc telluride (ZnTe:Cu) back contact for the Group 39 samples was reduced from one hundred (100) nanometers (nm) to fifty (50) nanometers (nm), the deposition temperature was increased, and other parameters were adjusted.

(83) The table **500** provides a baseline (row **511**) with zero percent (0%) hydrogen gas (H.sub.2), first target gas mix (row **513**) with point one percent (0.1%) hydrogen gas (H.sub.2), and second target gas mix (row **515**) with one percent (1.0%) hydrogen gas (H.sub.2). The graph **501** depicts average PCE data (y-axis **520**) via diagrams **531**, **533**, **535** respective to the table **500** for the three rows **511**, **513**, **515** of table **500**. The graph **502** depicts average voltage data (y-axis **520**) via diagrams **551**, **553**, **555** respective to the table **500** for the three rows **511**, **513**, **515** of table **500**. As seen in the FIG. **5**, the Group 39 samples with the adjusted parameters demonstrates both positive improvements based on hydrogen gas (H.sub.2) content in both electrical and optical properties of the completed solar cells.

(84) According to one or more embodiments, the Group 39 samples indicate that transmission in a visible spectrum (e.g., approximately two hundred fifty (250) nanometers (nm) to approximately eight hundred fifty (850) nanometers (nm)) increases as observed by the Thesis 1996 and Faulkner 2016, but also decreases minimally in the near infrared spectrum (eight hundred fifty (850) nanometers (nm) to one thousand two hundred (1200) nanometers (nm)). The Group 29 samples and the Group 39 samples also indicate that there is an optimal hydrogen gas (H.sub.2) concentration, above which, cells become non-functional while the optical properties continue to improve. Thus, the addition of the hydrogen gas (H.sub.2) to the argon (Ar) is not a simple process as this addition can yield non-functioning solar cells. The concentration of hydrogen gas (H.sub.2) present must be controlled and tuned specifically, along with other deposition parameters (e.g., thickness and temperature) to optimize a device. And these optimizations must be performed on a fully completed device as the thin film optimizations and corresponding results are unrelated and provide in incorrect conclusions.

(85) Additionally, data across the Group 29 samples and the Group 39 samples indicate how deposition parameters (for all sections) can influence a change in electrical and optical properties of that the hydrogen gas (H.sub.2) concentration has on a device. The deposition parameters include, but are not limited to, a copper-doped zinc telluride (ZnTe:Cu) back contact thickness, a copper-doped zinc telluride (ZnTe:Cu) back contact deposition temperature, a selenium concentration in a cadmium selenide telluride (CdSeTe) first transmissive absorber section, a cadmium selenide telluride/cadmium telluride (CdSeTe/CdTe) deposition temperature, a cadmium selenide telluride/cadmium telluride (CdSeTe/CdTe) post deposition annealing time, a cadmium selenide telluride/cadmium telluride (CdSeTe/CdTe) post deposition temperature, a MZO deposition temperature, and a cadmium chloride (CdCl2) post-deposition treatment temperature. Accordingly, the incorporation of hydrogen gas (H.sub.2) into a cadmium telluride (CdTe) based solar cell with a copper-doped zinc telluride (ZnTe:Cu) back contact requires testing, evaluation, and tuning of a 9-way parameter matrix, which at one set of parameters creates a totally non-functional device and at certain parameters creates and optimized electrical and optical device (showing unexpected results).

(86) FIG. **6** depicts a graph **600** according to one or more embodiments. The graph **600** depicts average transmissivity (x-axis **602**) of a copper-doped zinc telluride (ZnTe:Cu) back contact for three samples **621**, **623**, **624**. The graph key **630** identifies the bars as a maximum of a transmissivity percentage average **632** for two hundred fifty (250) nanometers (nm) to approximately eight hundred fifty (850) nanometers (nm) and a maximum of a transmissivity percentage average **634** for eight hundred fifty (850) nanometers (nm) to approximately one thousand two hundred (1200) nanometers (nm).

(87) FIG. **7** depicts a graph **700** according to one or more embodiments. The graph **700**. The graph **700** depicts average transmissivity (x-axis **702**) of a copper-doped zinc telluride (ZnTe:Cu) back contact for three samples **721**, **723**, **724**. The graph key **730** identifies the bars as a maximum of a transmissivity percentage average **732** for two hundred fifty (250) nanometers (nm) to approximately eight hundred fifty (850) nanometers (nm) and a maximum of a transmissivity

percentage average **734** for eight hundred fifty (850) nanometers (nm) to approximately one thousand two hundred (1200) nanometers (nm). The graph key **730** also identifies the bars as an average of a transmissivity percentage average **736** for two hundred fifty (250) nanometers (nm) to approximately eight hundred fifty (850) nanometers (nm) and an average of a transmissivity percentage average **738** for eight hundred fifty (850) nanometers (nm) to approximately one thousand two hundred (1200) nanometers (nm).

(88) FIG. **8** depicts an example deposition system **800** and method **801** according to one or more embodiments. The method **801** provides a calibrating of the example deposition system **800** to generate a plurality of time and temperature offsets. Note that the system and methods herein utilize the plurality of time and temperature offsets to achieve calibrated surface temperatures for manufacturing the transmissive cadmium (Cd) alloy solar cell. For ease of explanation, items and elements of the that are similar to previous Figures are reused and not reintroduced. Note that while one item is shown in the example deposition system **800**, any individual item can represent more than one of that item. FIG. **8** includes a surface **806** of a glass **807** on which one or more sensing devices, e.g., one or more thermocouples **810** and one or more temperature indicating stickers **820**.

(89) The method **801** begins at block **840**, where a glass is mounted in the vacuum chamber **303**.

(90) At block **845**, the or more sensing devices are added on the surface **806** of the glass **807**. According to one or more embodiments, the one or more thermocouples **810** are placed in contact with the surface **306**. For example, a spring rod is used to hold the thermocouple **810**. According to one or more embodiments, the one or more temperature indicating stickers **820** are placed on the surface **306**. For example, a spring rod is used to hold the thermocouple **810**.

(91) At block **850**, the substrate holder system **310** is set to a first set point temperature. The first set point temperature can include a value of one hundred fifty (150) degrees Celsius (° C.).

(92) At block **855**, a surface temperature on the surface **306** is continuously detected until a normalization. Normalization includes when the surface temperature has stabilized at a value.

(93) At block **860**, a time and a surface temperature offset is recorded at the normalization.

(94) At block **870**, the method **801** includes incrementing the first set point temperature to a next set point temperature. The first temperature is repeatedly incremented to the next set point temperature in twenty-five (25) degrees Celsius (° C.) increments. Then, the method **801** loops back to block **850** as shown by arrow **880**. Accordingly, the substrate holder system **310** is set to the next set point temperature, a next surface temperature is detected at a next normalization respective to the next set point temperature, and recording a next time and a next surface temperature offset respective to the next set point temperature until the substrate holder system **310** is set to a last set point temperature. The last set point temperature can include a value at or approximately at six hundred (600) degrees Celsius (° C.). Thus, the method **901** loops through substrate holder system **310** settings from cold to hot measure output of the thermocouple **810** and/or the stickers **920** until normalize and then increment. Note that a time to normalize increases and offset between setting and surface increase.

(95) At block **850**, the plurality of time and temperature offsets are generated. The plurality of time and temperature offsets are generated for the example deposition system **800** from each recording at each increment from the first set point temperature to the last set point temperature.

(96) FIG. **9** depicts a graph **900** according to one or more embodiments. The graph **900** provides visual representation of a temperature calibration according to any the method **801** of FIG. **8**. Plots of the graph **900** are depicted with respect to time (x-axis **905**), temperate degrees Celsius (° C.) (y-axis **906**), and rate of change degrees Celsius (° C.) (y-axis **907**). Graph key **910** identifies the plots of the graph **900**.

(97) According to the method **801**, the substrate holder system **310** is set to two hundred (200) degrees Celsius (° C.). Plot **921** shows an operation of the substrate holder system **310** as it heats from zero (0) degrees Celsius (° C.) to the two hundred (200) degrees Celsius (° C.) set point over approximately ten (10) minutes). Plot **922** shows temperature of the surface **806** of the glass **807** as it is heated by the substrate holder system **310** according to plot **921**, and plot **923** shows a corresponding the rate of temperature change. According to the method **801**, the set point of the substrate holder system **310** is incremented the next set point temperatures, i.e., three hundred (300) degrees Celsius (° C.) showing plots **934**, **935**, **936**, four hundred (400) degrees Celsius (° C.) showing plots **947**, **948**, **949**, and five hundred (500) degrees Celsius (° C.) showing plots **951**, **952**, **953**. Thus, the last set point temperature is five hundred (500) degrees Celsius (° C.).

(98) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one more other features, integers, steps, operations, element components, and/or groups thereof. It will be further understood that the terms “about” and/or “approximately,” when used in this specification, specify a degree of variation from a stated value that is can be detected and can be determined. For example, the degree of variation can be a ±1% from the stated value. The “optimized” and “optimization” terms (as well as the “high” term) reference changing a value of a particular property (in contemplation of other properties) over conventional technologies, scholarship, and research to achieve and comparatively improve commercially viable electrical performance with transmissivity and manufacturability.

(99) Although features and elements are described above in particular combinations, one of ordinary skill in the art will appreciate that each feature or element can be used alone or in any combination with the other features and elements. The

drawings, graphs, flowcharts, and diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of devices, apparatuses, and systems, and with respect to one or more methods, according to various embodiments. In this regard, each block in the flowchart or block diagrams may represent a module, a segment, a layer, a device, or portion of thereof. In some alternative implementations, the functions noted in the blocks may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved.
(100) The descriptions of the various embodiments herein have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A method of manufacturing a cadmium (Cd) alloy transmissive solar cell, the method comprising: pumping, by a pumping system, a vacuum chamber to a base pressure; pumping, by the pumping system, the vacuum chamber to raise the base pressure to a sputtering pressure; providing, by one or more mass flow controllers, into the vacuum chamber a first gas at a rate that balances a flow of the first gas in and out of the vacuum chamber with respect to the sputtering pressure; heating a surface of a partially manufactured cadmium (Cd) alloy transmissive solar cell within the vacuum chamber to a calibrated deposition temperature; providing, by the one or more mass flow controllers, into the vacuum chamber a second gas comprising at least a hydrogen gas (H.sub.2) at a proportional rate to achieve a target gas mix while maintaining the sputtering pressure by separately running the pumping system; and depositing a target material onto the surface to form a back contact section of the cadmium (Cd) alloy transmissive solar cell.
2. The method of claim 1, wherein the target gas mix comprise the hydrogen gas (H.sub.2) at or below 1.00%.
3. The method of claim 1, wherein the target gas mix comprises the hydrogen gas (H.sub.2) at or approximately at 0.1%.
4. The method of claim 1, wherein the hydrogen gas (H.sub.2) comprises a preblended hydrogen gas mix.
5. The method of claim 1, wherein the hydrogen gas (H.sub.2) and the first gas are provided across a plurality of phases to achieve the target gas mix.
6. The method of claim 1, wherein the target material comprises a copper-doped zinc telluride (ZnTe:Cu).
7. The method of claim 1, wherein the target material is deposited in a superstrate configuration onto the surface of a transmissive absorber section of the partially manufactured cadmium (Cd) alloy transmissive solar cell.
8. The method of claim 1, wherein the target material is deposited in a substrate configuration onto the surface of a back electrode section of the partially manufactured cadmium (Cd) alloy transmissive solar cell.
9. The method of claim 1, wherein the target material is deposited onto the surface at a rate of one (1) to five (5) angstroms per second to a section thickness of twenty (20) nanometers (nm) to one hundred and fifty (150) nanometers (nm).
10. The method of claim 1, wherein a deposition rate of the target material is controlled by setting a power supply of a sputter gun to a calibrated power density and performing the depositing the target material for a specified time to achieve a section thickness of twenty (20) nanometers (nm) to one hundred and fifty (150) nanometers (nm).
11. The method of claim 1, wherein a substrate holder system is set to a set point temperature to heat the surface to a calibrated deposition temperature along a range of two hundred (200) degrees Celsius (° C.) to six hundred (600) degrees Celsius (° C.).
12. The method of claim 1, wherein a substrate holder system is set to a set point temperature to heat the surface to a calibrated deposition temperature at or approximately at three hundred (300) degrees Celsius (° C.).
13. The method of claim 1, further comprising: depositing one or more substrate section materials; or depositing one or more conductive section materials.
14. The method of claim 1, further comprising: operating one or more shutters to control the deposition of the target material while adjusting the target gas mix.
15. The method of claim 1, wherein the first gas comprises argon gas (Ar).
16. The method of claim 1, further comprising: depositing one or more first transmissive absorber section materials.
17. The method of claim 16, wherein a substrate holder system is set to a set point temperature to heat a deposition surface of the partially manufactured cadmium (Cd) alloy transmissive to a calibrated first transmissive absorber deposition temperature along a range of two hundred (200) degrees Celsius (° C.) to six hundred (600) degrees Celsius (° C.).
18. The method of claim 17, wherein the one or more first transmissive absorber section materials comprise cadmium selenide telluride (CdSeTe) with a selenium concentration of 50% or more.
19. The method of claim 16, further comprising: depositing one or more second transmissive absorber section materials comprising a cadmium (Cd) alloy comprises cadmium selenide telluride (CdSeTe) or cadmium selenide (CdSe).
20. The method of claim 19, wherein a substrate holder system is set to a set point temperature to heat a deposition surface of the partially manufactured cadmium (Cd) alloy transmissive solar cell to a calibrated absorber temperature

along a range of one hundred (100) degrees Celsius (° C.) to three hundred fifty (350) degrees Celsius (° C.).
21. The method of claim 19, further comprising: performing a post deposition annealing of the one or more second transmissive absorber section materials at an annealing time and an annealing temperature.
22. The method of claim 19, further comprising: performing a cadmium chloride ($CdCl_2$) post-deposition treatment of the one or more second transmissive absorber section materials at a treatment temperature.
23. The method of claim 1, further comprising: depositing one or more materials for a back electrode section of the partially manufactured cadmium (Cd) alloy transmissive solar cell.
24. The method of claim 1, wherein the second gas is turned off during a pre-sputter phase comprising raising the base pressure to a sputtering pressure and one or more shutters are in a closed position.
25. The method of claim 24, wherein the second gas is turned on when the pre-sputter phase concludes with the base pressure raised to the sputtering pressure and the one or more shutters are moved to an open position.
26. The method of claim 1, wherein during the depositing of the target material a first shutter over the target material is opened while the vacuum chamber stabilizes and a second shutter over the cadmium (Cd) alloy transmissive solar cell is opened after the vacuum chamber stabilizes.